Tsaniya faridah Maulida
211331031
1A

No. :

Latihan Soal - Soal Resistor

1. a. kuning, ungu, emas, perak

   4 7 $\times 10^{-1}$ 10% = 4,7 Ω ± 10%

   b. Orange, putih, kuning, emas

   39 $\times 10^{4}$ ± 5% = 390 kΩ ± 5%

   c. Coklat, hitam, merah, emas

   10 $\times 10^{2}$ ± 5% = 1 kΩ ± 5%

   d. Coklat, hijau, hijau, tanpa warna

   15 $\times 10^{5}$ ± 20% = 1,5 MΩ ± 20%

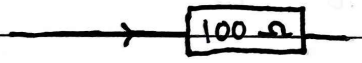
   e. Hijau, biru, hitam, merah, coklat

   560 $\times 10^{2}$ ± 1% = 56 kΩ ± 1%

   F. Coklat, abu-abu, hitam, hitam, Coklat

   180 $\times 10^{0}$ ± 1% = 180 Ω ± 1%

2. Hitunglah daya yang disisipkan pada masing-masing tahanan dengan kondisi tahanan di bawah ini :

   a. 

   100mA

   $P = I^2 . R$

   $= (0,1)^2 . 100$

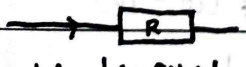
   = 1 Watt

   b. 

   |← 5V →|

   $P = \frac{V^2}{R}$

   $= \frac{(5)^2}{1k} = 0,025$ Watt

   c. 

   1A |← 5V →|

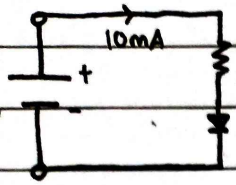
   P : V . I

   : 5 . 1

   : 5 Watt

No. :

3. Tentukan nilai resistor pada rangkaian ini yang sesuai dengan nilai standart resistor E12.

a) 

V = 5V

I = 10 mA = 0,01 A

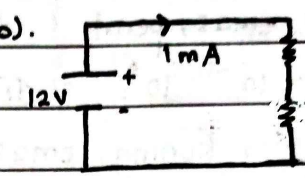
$V_S = 5V$

$V_S = V_{led} + V_{RS}$

$5 = 2 + V_{RS}$

$V_{RS} = 3V$

$R_S = \frac{V_{RS}}{I} = \frac{3}{0,01} = 300\Omega$

b). 

V = 12V

I = 1mA

= 0,001 A

$R_1 = \frac{V_{R1}}{I} = \frac{7}{0,001} = 7k\Omega$

$R_2 = \frac{V_{R2}}{I} = \frac{5}{0,001} = 5k\Omega$

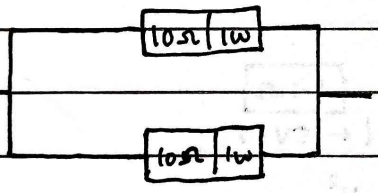
4. Hitung besar tahanan total dan daya total bila resistor 10Ω / 1w disusun secara :

a. Seri 

R total : $R_1 + R_2$

$= 10\Omega + 10\Omega = 20\Omega$

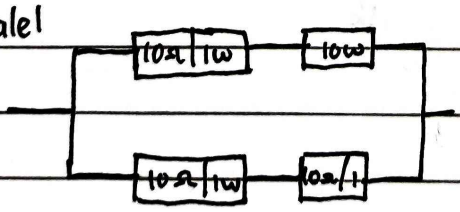
P : 2 watt

b. Paralel 

R total : $\frac{1}{10} + \frac{1}{10} = \frac{2}{10}$

r total : $\frac{10}{2} = 5\Omega$

P = 0,5 watt

c. Seri - Paralel 

R total = $\frac{(10+10) + (10+10)}{(10+10) + (10+10)}$

$= \frac{400}{40} = 10\Omega$

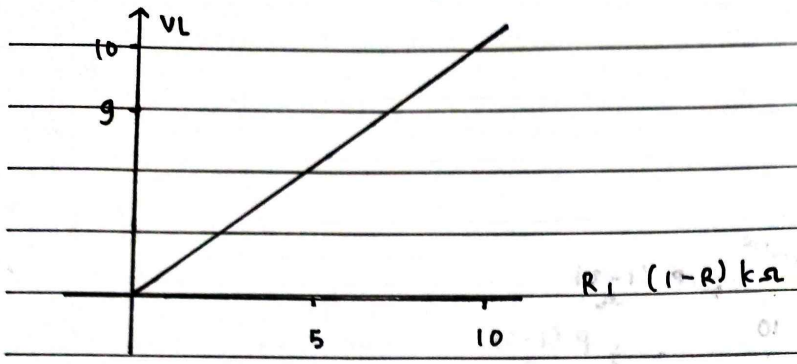
P = 1 watt

KIKY

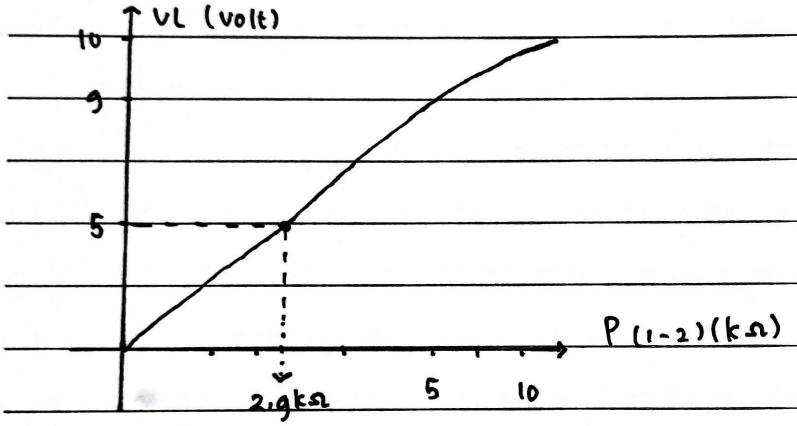
No. :

5. Gambarkan grafik $V_L = f(P_{1-2})$ untuk kondisi :

a) $R_L = \infty$ (tak hingga)

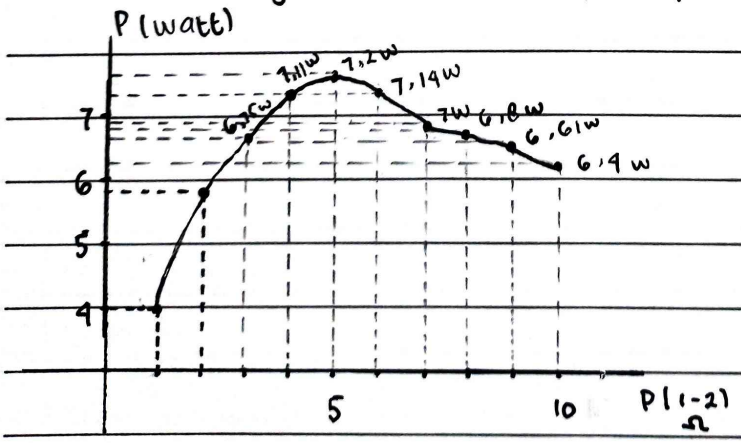


b). $R_L = 5\,k\Omega$



6. Nilai resistor variabel berubah dari $0 - 10\,\Omega$ [P_{1-2}]. Gambarkan pada diagram yang sama:

a. Grafik daya yg disisipasikan pada potensiometer $P_{(pot)} = f[P_{1-2}]$



$$I = \frac{V}{R_t} = \frac{12}{5+10} = 0,8\,A$$

$$P = I^2 \cdot R$$

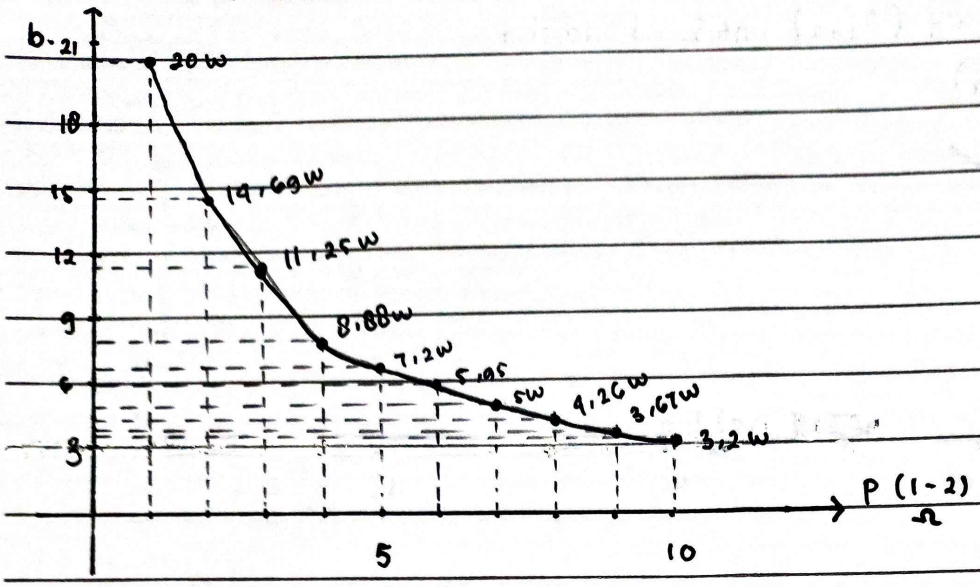
$$= 0,8^2 \cdot 10$$

$$= 6,4\,W$$

$$I = \frac{V}{R_t} \qquad P = I^2 \times R$$

KIKY

No. :



7. R_{LDR} = 300 Ω (terang)

V_o = 5V

$V_s = V_{R_1} + V_{LDR}$

$12 = 5 + V_{LDR}$

$V_{LDR} = 12 - 5 = 7V$

$$I = \frac{V_{LDR}}{R} = \frac{7}{300} = 23,34\ mA$$

$$R_1 = \frac{V_{R_1}}{I} = \frac{5}{23,4 \cdot 10^{-3}} = \frac{5000}{23,34} = \approx 214\ \Omega$$